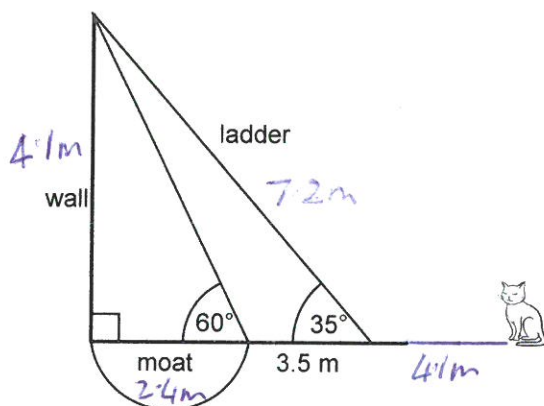


Calculator-assumed, 1 A4 page notes allowed. Formula Sheet provided.

Time: 50 minutes

QUESTION 1 [7 marks – 2, 2, 3]

A wooden ladder is leaning up against a castle wall. Under the ladder, and in-front of the wall, is a moat 2.4 metres wide which is full of water, as shown in the diagram below.



- a) Show working to determine that the length of the ladder is approximately 7.2 metres.

$$\cos 35 = \frac{2.4 + 3.5}{L}$$

$$L = \frac{5.9}{\cos 35} = 7.2 \text{ m}$$

✓ Uses cosine ratio

✓ Calculates length

- b) Determine the height of the castle wall.

$$\tan 35 = \frac{h}{5.9}$$

$$h = 5.9 \tan 35$$

$$= 4.1 \text{ metres}$$

$$\sin 35 = \frac{h}{7.2}$$

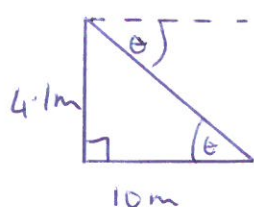
$$h = 7.2 \sin 35$$

$$= 4.1 \text{ m}$$

✓ Use tan or sine ratio or Pythagoras Theorem.

✓ Calculates height

- c) If a bird landed on top of the castle wall and saw a cat standing 10 metres away from the base of the castle, what angle of depression, to the nearest degree, would the bird be looking at towards the cat?



$$\tan \theta = \frac{4.1}{10}$$

$$\theta = \tan^{-1} \left(\frac{4.1}{10} \right)$$

$$= 22.44$$

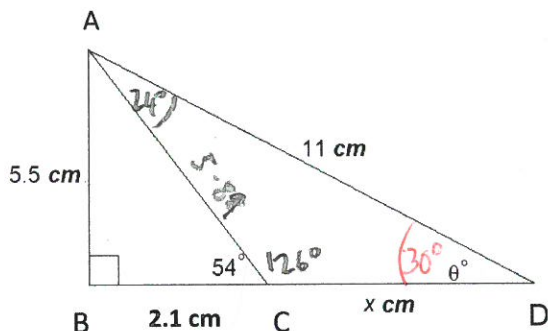
$$\sim 22^\circ$$

✓ Identifies angle of depression

✓ Uses tan ratio

✓ Calculates angle to nearest degree.

QUESTION 2 [7 marks – 2, 3, 2]



- a) Find the size of angle D.

$$\sin \theta = \frac{5.5}{11}$$

$$\theta = \sin^{-1}\left(\frac{5.5}{11}\right) = 30^\circ$$

✓ Uses sine ratio

✓ Calculates angle

- b) Find the length of x cm.

$$\tan 30^\circ = \frac{5.5}{(x+2.1)}$$

$$(x+2.1) = \frac{5.5}{\tan 30} = 9.526$$

$$\therefore x = 7.43 \text{ cm}$$

✓ Uses tan ratio

✓ Calculates base

✓ Subtracts to calculate x cm.

- c) Find the area of $\triangle ACD$.

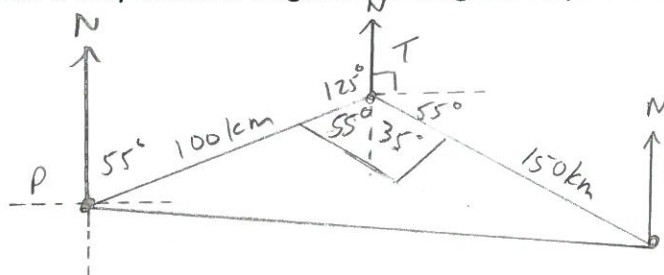
$$A = \frac{1}{2} \times 11 \times 7.43 \times \sin 30 = 20.42 \text{ cm}^2$$

✓ Substitutes into Area formula
✓ Calculates Area.

QUESTION 3 [5 marks – 2, 1, 2]

A ship sails 100 km from a port Pianto (P) on a bearing of 055°T arriving at Taranto (T). It then travels 150 km on a bearing of 145°T until it reaches its destination at port Santo (S).

- a) Draw a fully labelled diagram showing the ship's travels.



✓ Shows correct bearings: $55^\circ, 145^\circ \text{T}$.

✓ Labels all angle and distances provided.

- b) What true bearing is Pianto from Taranto?

$$180 + 55 = 235^\circ \text{T}$$

✓ States true bearing

- a) Calculate the distance travelled if the ship takes a return journey directly from Santo to Pianto.

$$PS = \sqrt{100^2 + 150^2} = 180.28 \text{ km}$$

✓ Calculates using Pythagoras Theorem
✓ Calculates distance.

QUESTION 4 [6 marks – 3, 1, 2]

The construction company hired to build the Senior School Hub regularly tested cubes of the concrete produced to ensure its strength and durability as too much or not enough cement would impact the quality. The amount of cement in $1\,000\text{cm}^3$ of concrete tested was normally distributed with a mean of 420g and a standard deviation of 25g.

a) What range of cement quantities were needed to make:

i) 68% of concrete to an acceptable level?

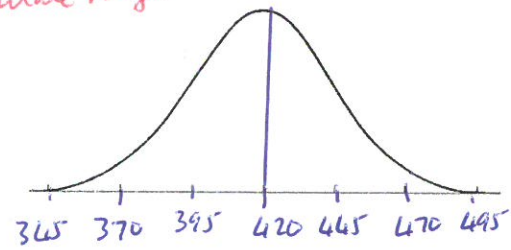
395g to 445g

ii) 95% of concrete to an acceptable level?

370g to 470g

iii) 99.7% of concrete to an acceptable level?

365g to 495g

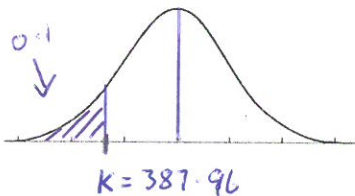


b) Find the probability that a concrete cube chosen at random had less than 380 grams of cement.

$$P(X < 380) = 0.0548$$

✓ Finds probability to left of p

c) If it was decided that the lowest 10% of cement quantities was not safe enough for the building, then how many grams of cement would be the lowest amount acceptable for the concrete? Represent this on the diagram below.



$$P(X \leq k) = 0.1$$

$$k = 387.96g$$

INV Dist: Left

$\sigma = 25$

$\mu = 420$

✓ Show graph with 0.1 and k w 0.9

✓ Calculates lowest value acceptable

QUESTION 5 [5 marks – 1, 1, 3]

In the Senior School cafeteria the types and numbers of cold drinks leftover at the end of recess was recorded by a group of students. Students found their options were Juice(J), Milk Drinks(M), Iced Tea(T) and Water(W). They recorded what they saw left in the fridge below:

T J W T W M M M J W T T J J T

a) Circle 2 words below that describe the type of data collected in their research.

Categorical

Numerical

Nominal

Ordinal

Discrete

Continuous

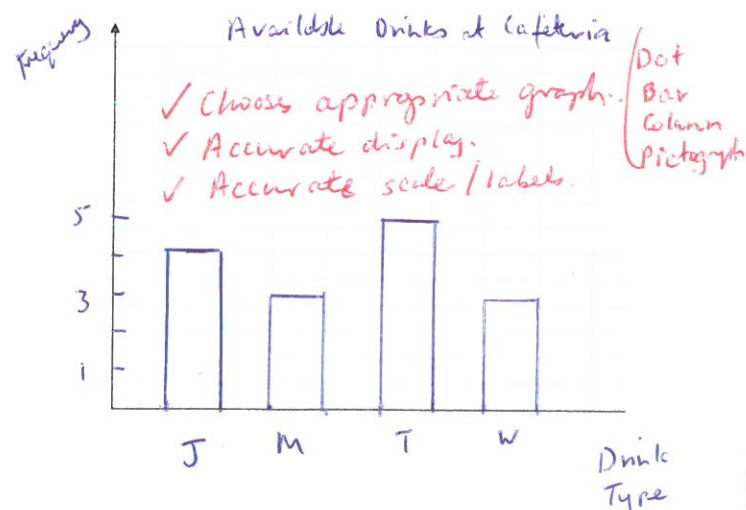
✓ Identifies type of data.

b) Complete the frequency table below:

c) Display the data in an appropriate graph.

Drink Type	Tally	Frequency
Juice(J)		4
Milk(M)		3
Iced Tea(T)		5
Water(W)		3
TOTAL		15

✓ Completes tally and frequency



QUESTION 6 [8 marks – 1, 1, 2, 2, 2]

The results obtained by 15 students in a Maths test are displayed below in the stem and leaf graph. The Maths test was out of 50 marks.

Maths Test Results

1	2	6
2	0	0 4 4 6 8 8 8
3	0	0 4 6
4	0	

Key: 1 2 = 12

- a) Calculate the mean of the test results.

$$\bar{x} = 26.4$$

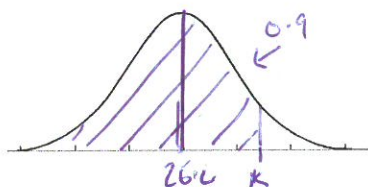
✓ Calculates mean

- b) Calculate the standard deviation of the test results.

$$\sigma_x = 7.2$$

✓ Calculates standard deviation

- c) If the Maths Test results are normally distributed, and a student wanted to be at the 90th quantile for this test, what test result did they need to attain?



$$P(X \leq k) = 0.9$$

$$k = 35.63$$

$$\sim 36$$

✓ Identifies $\leq 90\%$

✓ Calculates result.

- d) If Luke's result was 34 in the test, how many standard deviations is he above or below the mean?

$$z = \frac{34 - 26.4}{7.2}$$

$$= 1.056 \text{ st. dev. above}$$

✓ Uses standard score

✓ Calculates no. of st. dev. above.

- e) If Amy was absent for the test, but it was known she was 1.6 standard deviations above the mean for the previous test, what would be her expected result for this test?

$$1.6 = \frac{x - 26.4}{7.2}$$

$$x = 37.92$$

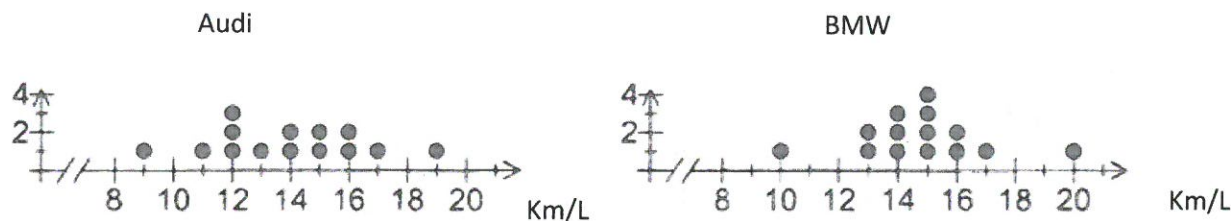
$$\sim 38 \text{ marks.}$$

✓ Uses z score formula

✓ Calculates expected result.

QUESTION 8 [9 marks – 1, 1, 1, 2, 2, 2]

RAC car magazine conducted some research to compare the fuel efficiency of two popular car brands. They chose to compare fuel economy by testing 14 cars of each brand. The dot plots below were created displaying the number of cars of each company achieving particular whole kilometres per Litre of fuel for Audi and BMW cars.



- a) Circle 2 words below that describe the type of data displayed in this research. *✓ Circle type of data*

Categorical Numerical Nominal Ordinal Discrete Continuous

- b) If RAC compare car manufacturers by only considering the car with the very highest kilometres per Litre fuel, which manufacturer is the most efficient? State your evidence.

BMW 20 km/Litre

✓ Identifies highest with evidence.

- c) If RAC only considered the mode of fuel consumption, which car manufacturer would they choose as being the most efficient? State your evidence.

BMW 15 km/Litre

✓ Identifies highest mode with evidence.

- d) If RAC only considered the mean kilometres per Litre, which car manufacturer would be the most efficient? Justify your answer.

$$\bar{x}_A = \frac{195}{14} = 13.93 \quad \bar{x}_B = \frac{207}{14} = 14.79$$

✓ Calculates both means.

✓ Chooses BMW.

∴ BMW as higher mean (14.79)

- e) Identify which car manufacturer's efficiency appears the most consistent. Explain your reasoning using relevant statistics.

BMW, as st dev 2.177 compared to Audi st. dev 2.52

BMW most consistent, varies less.

✓ Choose BMW

✓ Explains with st. deviations.

- f) Briefly compare the shape of the distributions of Audi and BMW fuel efficiency.

Both are fairly symmetrical

BMW is clustered more tightly (13-16 km/L) than

Audi (11-17 km/L).

Both have gaps at top and bottom ends of data.

✓ Compares symmetry

✓ Compares clusters or gaps.

QUESTION 7 [12 marks – 4, 2, 3, 3]

A group of 16 students completed as many sit ups as possible in 2 minutes prior to and after completing 6 weeks of fitness training. The number of sit ups completed were recorded below:

BEFORE:

23 22 33 19 19 15 26 17
18 31 24 20 29 45 28 22

AFTER:

30 26 45 32 25 26 42 36
38 21 26 34 37 37 7 34

- a) Record the following statistics for each group of data:

BEFORE:

Median: 22.5

AFTER:

Median: 33

Interquartile Range: $28.5 - 19 = 9.5$

Interquartile Range: $37 - 26 = 11$

Range: $45 - 15 = 30$

Range: $45 - 7 = 38$

✓ Calculates a Median

✓ Calculates an IQR

✓ Calculates a Range

✓ Calculates all correctly.

- b) Show how to calculate if there are any outliers in the data recorded after the 6 weeks of fitness training and state them, if any.

AFTER:

$$26 - (1.5 \times 11) = 9.5$$

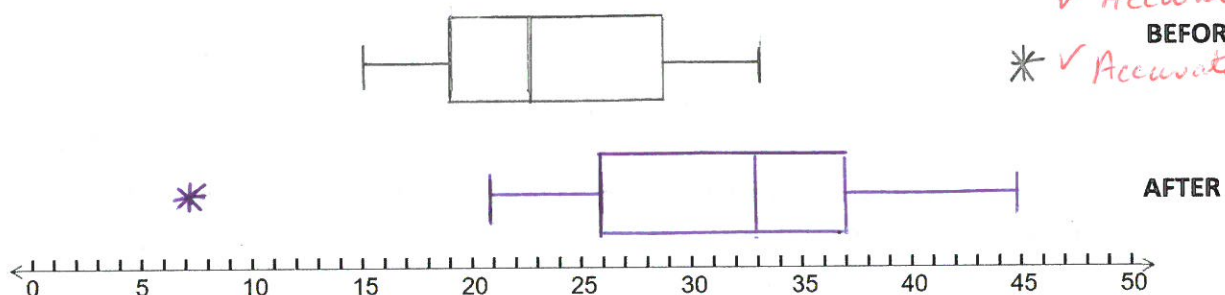
$$37 + (1.5 \times 11) = 53.5$$

Yes outlier of 7.

✓ Show calculations for outliers

✓ States outlier

- c) Represent the set of data as a parallel box plot in the position indicated, clearly showing outliers if appropriate.



✓ Accurate box
✓ Accurate whiskers

* ✓ Accurate outlier

- d) Referring to your previous answers briefly compare location, and spread of the two groups of data and comment on any reason why the class may have attained these results.

Median AFTER is 11 sit ups higher than median BEFORE, indicating better results after.

IQR AFTER 1.5 sit ups higher than IQR BEFORE of only 9.5 sit ups indicating greater spread after fitness training occurred. Overall fitness has improved with 6 weeks of training, but some students did not improve much, including outlier, who may have been absent.

✓ Compares MEDIAN

✓ Compares Spread

✓ Comments appropriately